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Population model

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Photo of red-bearded bee-eater by JJ Harrison.

Interacting populations

Choose your own set of populations: animals, plants, bacteria... and model the size of the populations. One population might feed on the other, but you can also choose two populations that compete for the same resources or that the two species benefit from each other.

Population model

The number of members in the first population is called $u(t)$ and the number of members of the second population $v(t)$.

$$\begin{aligned}\frac{du}{dt} &= \alpha \left(1 - \frac{u(t)}{k} \pm \frac{\beta v(t)}{1 + \gamma u(t)} \right) u(t) \\ \frac{dv}{dt} &= \left(-\delta \pm \frac{\beta u(t)}{1 + \gamma u(t)} \right) v(t)\end{aligned}$$

Tasks for the project in Module 4

Finish the modelling for the full basic model. First a repeat of the optional tasks in module 3; if you have not done them there, do them here.

1. Implement Euler's method for the system, with $\beta > 0$, and start with $\gamma = 0$. If you have decided on different stepsizes for the calculations of the separate $v(t)$ and $u(t)$, for the system you need to calculate with the minimum of the two stepsizes. If in doubt check the errors again.
2. Find a reasonable value for β . There are different ways for finding a value:
 - A quantitative argument, such as we gave for the gouramis.
 - Numerical simulation experiments and see whether the resulting interaction is plausible.
 - Analytical calculation of the equilibrium points, especially the one with both populations present. Are the numbers of the populations in that equilibrium in the right proportion? It is probably wise, to start with $\gamma = 0$.
 - Calculation of the eigenvalues of the Jacobian matrix in the end state equilibrium point. If the solution oscillates, determine the period of the oscillations and set β to a value such that the period of the oscillations is plausible.

Ideally, you use more methods to determine β . On the other hand, maybe you want to press on, and focus on other parts of the modelling process. Then you use just one method.

3. Start including $\gamma > 0$. Same options to choose a value as for β . Is the interaction of the two species modelled significantly better, or does it not make much difference?

New points for module 4:

4. Calculate the equilibrium points and their types for the parameters chosen.
5. Simulate the population sizes for longer times. Generate graphs of $u(t)$ and $v(t)$ versus time. Are the populations oscillating? What is then the period? Who is first, who lags behind? What is the phase difference? Are these values reasonable for your populations?
6. Generate graphs of the solutions in the phase plane: v versus u . Use different sets of initial conditions, and plot all solutions in the same phase plane.
7. Optional: investigate how the positions of the equilibrium points depend on the value of one or more of the parameters. You could even investigate whether the type of one of the equilibrium points changes with a different value of one of the parameters.

Extending your model

After you have written and submitted your first report, in module 5 you have to go beyond the basic model. You should start thinking about what you would like to do then. What specific problem about the two populations do you want to solve? That can be a pressing real-life problem, or you could investigate something you made up. What if ...?

So you could investigate what happens when there is some intervention: hunters may be allowed to shoot more or fewer of one of the species, extra food is provided, a third species is introduced,... An intervention can be at $t = 0$, or you could simulate it to take effect after a certain time by including an if statement: If $t \geq 50$: and then your intervention for example changes the derivatives. Or maybe you want to include different effects for days and nights or times of the year. Then your final modelling problem could be What is the influence of ...?

You may be dissatisfied with this model for your populations, and you may alter the terms, such that they model your populations better. Then your modelling problem is could be What is a mathematical model to predict ...?

You could also go more into the theoretical aspects of the model, as in question 7 above. Your modelling problem is then a more theoretical problem, interesting for others working with the same kind of models.

The extension becomes only relevant in module 5. Finish your work on the basic model first and write and submit the first report before June 27th.

After the pages for the other problems, you will find a page for questions you might have about the project.

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